Unit 1 · Session 1 · Thu, Oct 15, 2026

Introduction to Pharmacology — Session 1

5 parts · every fact cited to its slide or textbook page. About 160 minutes of focused study.

Study plan

20 min
Read the guide
One pass, top to bottom. Bold = what the lecture emphasized.

60 min
Diagrams & tables
Redraw each flow from memory; cover a table column and recall it.

35 min
Cram sheet
Hide answers and quiz yourself. Repeat until it's automatic.

40 min
Self-test
Answer before you look. Read every explanation, even when right.

5 min
Objectives check
Tick each objective only if you can teach it out loud.

HOW TO READ THIS GUIDE

- **Bold** = emphasized on the lecture slides
- TB textbook only · SG course study guide only
- S12 = slide 12 · S12 (notes) = speaker notes · SG U1-R3 = study guide item · p. 17 = Katzung page
- Open **Sidekick** (bottom right) to search, listen, drill flashcards, quiz, or chat about this lecture

The whole lecture in 6 sentences

1. Pharmacology studies how drugs interact with living systems: **pharmacokinetics** is what the body does to the drug (ADME), and **pharmacodynamics** is what the drug does to the body. S5; S6

2. Learn and prescribe by **generic name** — brand names can hide a shared ingredient and cause unintentional double-dosing. S7; S9

3. Drugs move from discovery through preclinical testing, **Phase I–III** trials, FDA review, and **Phase IV** postmarketing surveillance — where rare adverse effects are often first found. S12; S13; S12 (notes)

4. Drugs act on five target classes; a **competitive antagonist** shifts the dose-response curve right (same Emax), while a **noncompetitive/irreversible** antagonist lowers Emax. S20; S22 (notes)

5. **Potency** (EC_{50}) is the dose needed and **efficacy** (Emax) is the ceiling — efficacy matters more — and narrow-TI drugs (**warfarin, lithium, digoxin, phenytoin**) need careful dosing and blood levels. S30; S34; S38

6. Oral drugs face **first-pass** hepatic metabolism that lowers bioavailability (IV = 100%), and distribution depends on protein binding, lipid solubility, perfusion, and barriers like the BBB. S50; S51; S52; S53

Learning objectives

IO = syllabus unit objective · SO = this session's objectives slide.

IO1.1 Define and describe fundamental pharmacologic concepts, including pharmacokinetics, pharmacodynamics, drug-receptor interactions, dose-response relationships, therapeutic index, half-life, and steady state.

⚠ Half-life and steady state are taught in Session 2 (L02)

Part 1 · The Language of Pharmacology
Part 3 · Pharmacodynamics: Targets & Receptors
Part 4 · Dose-Response Relationships

IO1.2 Describe the processes of drug absorption, distribution, metabolism, and excretion and factors that influence these processes.

⚠ This session: absorption + distribution; metabolism + excretion detail in Session 2 (L02)

Part 5 · Pharmacokinetics: ADME — Absorption & Distribution

IO1.3 Explain how patient-specific factors, including age, pregnancy, renal and hepatic function, and pharmacogenetic variation, may affect medication response.

⚠ Barely covered this session; pharmacogenomics and renal dosing in Session 2 (L02)

Part 5 · Pharmacokinetics: ADME — Absorption & Distribution

IO1.4 Identify common types and mechanisms of adverse drug reactions, drug-drug interactions, and drug-disease interactions.

⚠ Only ADR definition + adverse-event reporting this session; drug interactions in Session 2 (L02)

Part 1 · The Language of Pharmacology
Part 2 · Drug Development & Regulation

IO1.5 Interpret common components of medication orders and prescriptions, including drug name, dose, route, frequency, and duration.

⚠ Only drug names (generic vs brand) and routes this session

Part 1 · The Language of Pharmacology
Part 5 · Pharmacokinetics: ADME — Absorption & Distribution

SO1 Define pharmacology, pharmacodynamics, and pharmacokinetics.

Part 1 · The Language of Pharmacology

SO2 Describe the major stages of drug development and FDA approval.

Part 2 · Drug Development & Regulation

SO3 Explain how drugs interact with receptors to produce an effect.

Part 3 · Pharmacodynamics: Targets & Receptors

SO4 Interpret a dose-response curve, including potency and efficacy.

Part 4 · Dose-Response Relationships

SO5 Explain absorption, distribution, metabolism, and excretion (ADME), including bioavailability and the first-pass effect.

Part 5 · Pharmacokinetics: ADME — Absorption & Distribution

ABOUT THIS LECTURE

- Session 1 of 2 on Introduction to Pharmacology (Thursday, October 15): language, drug development, pharmacodynamics, dose-response, and ADME through distribution. S1; S2 (notes)
- Session 2 (Thursday, October 22) continues with metabolism, excretion, PK calculations (half-life, steady state, clearance), pharmacogenomics, and the integrated clinical decision-making framework. S2; S56
- ADME is only half-finished after Session 1 — the lecture stopped after Distribution; Metabolism and Excretion pick up first in Session 2. S56 (notes)
- Required text: Katzung's Basic and Clinical Pharmacology, 16th Edition (Todd W. Vanderah · 2025 · McGraw-Hill Education · ISBN 1260463303 eBook) — Unit 1 reading: Chapters 1, 2, 3, 4, and 5. AccessMedicine: accessmedicine.mhmedical.com/book.aspx?bookid=3382 S3
- Recommended text: Pharmacotherapy Principles and Practice (Chisholm-Burns, Schwinghammer, Malone, Kolesar, Lee · 2022 · McGraw-Hill Education · Print ISBN 9781260460278 · eText ISBN 9781260460285) — use as supplemental support alongside Katzung's Chapters 1–5, not a substitute. S3; S3 (notes)
- **Original figures in this deck:** the lecture's Figures 1–5 (agonist & antagonist effects; sigmoidal dose-response curve; potency vs. efficacy; wide vs. narrow therapeutic index; ADME compartment model) are original diagrams generated by the course director using the AI tool Claude (Anthropic, 2026, Claude Sonnet 5) to illustrate standard, non-copyrighted pharmacologic concepts — not reproduced from Katzung or any other copyrighted source. **Reference:** Anthropic. (2026). Claude (Sonnet 5) [Large language model]. S57; S57 (notes)

SLIDE DISCREPANCIES

- **First-pass effect:** slides describe hepatic first-pass and say essentially every route except oral bypasses it. The textbook adds that drug can also be metabolized in the gut wall, and that the rectal route only partly avoids the liver (about 50%). S49 (notes); S51; p. 48; p. 49
- **Therapeutic index:** slides define TI = TD_{50} ÷ ED_{50}. The textbook notes this is measured in animals; for a patient, the therapeutic window (minimum therapeutic to minimum toxic dose) is more useful. S38; p. 38
- **Phase I population:** slides say healthy volunteers; the textbook adds that patients with the disease are used when significant toxicity is expected (cancer, AIDS). S13; p. 17
- **Phase II size:** slides say a few hundred patients; the textbook says 100–200. Exams may use either — know both. S13; p. 17

1 PART 1 · TOPIC

The Language of Pharmacology

SO1 IO1.1 IO1.4 IO1.5

Key definitions

- **Pharmacology:** the scientific study of how drugs interact with living systems — from the molecular level to the whole patient. S5
- **Drug (medication):** any chemical substance used to diagnose, treat, cure, or prevent disease, or to alter a physiologic process. S5
- **Pharmacologic effect:** a measurable biological change a drug produces (e.g., lower blood pressure). S5
- **Therapeutic effect:** the intended, beneficial pharmacologic effect for a given condition. S5
- **Adverse drug reaction (ADR):** an unintended and harmful response occurring at normal therapeutic doses. S5
- **Toxicity:** harmful effects that occur as the dose or exposure exceeds the therapeutic range. S5
- **Chemical name:** describes the exact molecular structure (rarely used in clinical practice). S7
- **Generic name:** the official, non-proprietary name assigned to the active drug — one generic name per drug, regardless of manufacturer. S7
- **Brand (trade) name:** the manufacturer's marketed name; the same generic drug may have several brand names, and brand names change between countries. S7
- TB **Prodrug:** an inactive precursor converted to the active drug in the body (usually by metabolism). p. 8
- TB **Poisons** are drugs with almost exclusively harmful effects — but "the dose makes the poison" (Paracelsus): any

How it works

- Pharmacokinetics = what the body does to the drug — ADME. S6; SG U1-R1
- Pharmacodynamics = what the drug does to the body — receptors & effect. S6; SG U1-R1
- Pharmacogenomics = how genetic variation shapes individual drug response. S6
- Clinical Pharmacology / Toxicology = applying PK/PD to real patients, and the study of poisoning. S6
- This map covers the whole lecture: PK and PD are the two biggest buckets, and nearly everything else (toxicology, pharmacogenomics, therapeutics) sits on top of them. S6 (notes)
- TB Drugs bind receptors by **covalent** bonds (strong, often irreversible), **electrostatic** bonds (most common) or **hydrophobic** bonds (weak). p. 4
- TB **Chirality:** over half of drugs are mirror-image pairs (enantiomers); one is usually much more potent, yet most are sold as 50:50 mixtures. p. 4; p. 5

substance can harm at the wrong dose. p. 3

Drugs named as examples

- Acetaminophen (generic) is sold as Tylenol and under many store brands. S7

CLINICAL

- **PA students should learn and prescribe by generic name — it is unambiguous, international, and avoids confusing a patient's multiple brand-name products for different drugs.** S7; SG U1-R2
- Different brand-name products (Tylenol, many combination cold/flu products) can share the same active generic ingredient. S9; SG U1-R2
- Prescribing and counseling by generic name avoids hiding duplicate therapy — the reason PAs default to generic names. S9; SG U1-R2
- If a patient says they take "Tylenol" and separately "acetaminophen," check for accidental double-dosing — this previews the acetaminophen toxicity discussion. S7 (notes)
- Precise vocabulary (generic names, therapeutic vs. adverse effects) protects patients. S56
- TB More than half of prescriptions in the USA are for generic drugs. p. 18
- TB Weak bonds need a precise fit, so they are **more selective**. Aspirin's covalent bond to platelet cyclooxygenase lasts until new platelets are made. p. 4

TRAP

- "Therapeutic effect" and "pharmacologic effect" are not synonyms — a drug can have pharmacologic effects that are not therapeutic (that is often exactly what an adverse effect is). S5 (notes)
- ADR vs. toxicity: an ADR happens at normal therapeutic doses; toxicity happens when dose or exposure exceeds the therapeutic range. S5
- Generic names do NOT vary by manufacturer (one per drug); brand names DO vary — and change between countries. S7

Chemical vs. generic vs. brand name

Name type	What it is	How many per drug	How to tell apart / clinical use
Chemical	Exact molecular structure	Not covered in materials	Rarely used in clinical practice
Generic	Official, non-proprietary name of the active drug	One — regardless of manufacturer	**Learn and prescribe by it** — unambiguous, international
Brand (trade)	Manufacturer's marketed name	Several possible; change between countries	Can hide a shared generic ingredient (e.g., Tylenol = acetaminophen)

How to tell them apart. Sources: S7; S9

Pharmacologic effect vs. therapeutic effect vs. ADR vs. toxicity

Term	Intended?	Harmful?	Dose
Pharmacologic effect	Not necessarily	Not necessarily	Any measurable biological change
Therapeutic effect	Yes — intended, beneficial	No	For a given condition
Adverse drug reaction (ADR)	No — unintended	Yes	Normal therapeutic doses
Toxicity	No	Yes	Dose or exposure exceeds the therapeutic range

How to tell them apart. Sources: S5; S5 (notes)

WHY IT HAPPENS

- Why prescribe by generic name? One drug has ONE generic name but can hide under SEVERAL brand names — so brand-name products can conceal duplicate therapy and cause double-dosing. S7; S9

MNEMONIC (MEMORY AID)

- Pharmaco**K**inetics = the body **K**eeps the drug moving (ADME). Pharmaco**D**ynamics = what the **D**rug **D**oes to the body. S6
- ADR = **A**t normal **D**oses, **R**eaction is harmful. Toxicity = **T**oo much (exceeds the therapeutic range). S5

HIGH-YIELD

- PK = what the body does to the drug (ADME); PD = what the drug does to the body (receptors & effect). S6; SG U1-R1
- ADR = unintended, harmful, at **normal therapeutic doses**; toxicity = dose/exposure **exceeds the therapeutic range**. S5
- Learn, prescribe, and counsel by **generic name** — brand names can hide a shared ingredient → double-dosing. S7; S9
- Pharmacology connects a drug's chemistry to its clinical effect. S56

2 PART 2 · TOPIC

Drug Development & Regulation

SO2 IO1.4

Key definitions

- **Efficacy:** how well a drug works under ideal, controlled trial conditions. S14
- **Effectiveness:** how well a drug works in everyday clinical practice, with real-world adherence and comorbidities. S14
- SG Drug interactions can also reduce real-world effectiveness below trial efficacy. SG U1-R4
- **Benefit vs. risk:** the central question regulators and prescribers must weigh for every approval and every prescription. S14
- **Adverse-event reporting:** ongoing, voluntary and mandatory reporting systems (e.g., FDA MedWatch) that catch safety signals after approval. S14
- TB **Double-blind:** neither subjects nor evaluators know who gets drug or placebo. **Crossover:** each subject gets every treatment in turn. p. 14; p. 15
- TB **IRB:** the committee at each trial site that approves the ethics of the study; participants give informed consent. p. 17

How it works

- Where new drugs come from — Natural products: plants, fungi, microbes, marine organisms. S11
- Synthetic & biotech: designed small molecules; recombinant proteins. S11
- Biologics: monoclonal antibodies, gene- and cell-based therapies. S11
- Drug repurposing: an existing approved drug found to treat a new condition. S11
- From discovery to patient: Discovery → Preclinical testing → Clinical trials → FDA review → Approval → Postmarket surveillance. S12
- Most candidate drugs fail somewhere along this pipeline — approval is the exception, not the rule. S12
- Phase I: safety, tolerability, dose-finding — a small group of healthy volunteers. S13; SG U1-R3
- Phase II: preliminary efficacy and side-effect profile — a few hundred patients with the target condition. S13; SG U1-R3
- Phase III: confirm efficacy and monitor adverse effects at scale — hundreds to thousands of patients, often multi-site. S13; SG U1-R3
- Phase IV: postmarketing surveillance after approval — the general prescribed population. S13; SG U1-R3
- Every approved drug has passed through a structured development and regulatory pipeline (Phases I–IV). S56
- TB Phase II has the **highest failure rate** — only about 25% of new drugs move on to Phase III. p. 17

- TB Phase III usually enrolls **thousands** of patients (Figure 1–6: 1000–6000), often double-blind and crossover, in settings like the drug's eventual use. p. 12; p. 17
- TB Less than one-third of drugs studied in humans survive clinical trials and reach the market. p. 14
- TB To see one case of a toxicity occurring 1 in 10,000, several hundred thousand patients may need exposure — so rare effects usually surface only in Phase IV, which has **no fixed duration**. p. 18
- TB Marketing requires a **New Drug Application (NDA)** after Phase III; human testing starts only after an **IND** (Investigational New Drug) application. p. 12; p. 17
- TB Strength of evidence: case reports < cohort and case-control studies < **randomized controlled trials** (the gold standard: randomization balances confounders). p. 15

Drugs named as examples

- FDA MedWatch — an adverse-event reporting system. S14
- Biologic DMARDs (MSK unit, later) are a direct example of the "biotechnology" drug source. S11 (notes)
- TB **Thalidomide** (sold in Europe as a "nontoxic" hypnotic) caused about 10,000 birth defects and led to the **Kefauver-Harris Amendments (1962)**, which required proof of efficacy as well as safety. p. 16
- TB The **Orphan Drug Amendment (1983)** gives incentives for drugs treating diseases with fewer than 200,000 US patients. p. 16

CLINICAL

- Postmarketing surveillance (Phase IV) is where rare adverse effects are often first detected — trial populations are too small and too short in duration to catch 1-in-10,000 events. S12 (notes)
- Case: a new medication improves symptoms in 70% of patients with a chronic, non-life-threatening condition, but postmarketing data show serious hepatotoxicity in about 1%. **Would you prescribe it — and does your answer depend on what else is available?** S15

- **Clinical pearl:** risk-benefit analysis is always comparative — a 1% serious-toxicity rate reads very differently if it is the only effective option versus if three safer alternatives already exist. S15
- Ask "compared to what?" rather than judging the number in isolation. S15 (notes)
- TB A patent lasts 20 years; after it expires, generics are approved through an **abbreviated new drug application (ANDA)** that requires bioequivalence data. p. 16; p. 18
- TB Placebo responses can occur in as many as 30–40% of subjects — the reason controlled, blinded trials are needed. p. 14
- TB Adverse events that are both **serious and unexpected** must be reported to the FDA within 15 days; clinicians can report through MedWatch. p. 18; p. 19
- TB Dietary supplements are regulated as food under DSHEA (1994), so they get no full FDA drug review. p. 3; p. 16

TRAP

- Efficacy vs. effectiveness is a favorite exam distinction: efficacy is the trial result; effectiveness is what happens once real patients with real adherence patterns use the drug. S14 (notes)
- Phase III — not Phase I, II, or IV — is the large pre-approval trial that confirms efficacy and monitors safety at scale. S17
- Phase III is what most people picture as "the" clinical trial, but regulatory decisions already begin shaping the drug long before that. S13 (notes)

Step by step

From discovery to patient S12; S13

Discovery → Preclinical testing → Clinical trials (Phases I–III) →
FDA review → Approval → Postmarket surveillance (Phase IV)

Clinical trial phases

Phase	Primary purpose	Typical population	How to tell apart
Phase I	Safety, tolerability, dose-finding	A small group of healthy volunteers	Only phase in **healthy volunteers**
Phase II	Preliminary efficacy and side-effect profile	A few hundred patients with the target condition	First look at whether it works
Phase III	Confirm efficacy and monitor adverse effects at scale	Hundreds to thousands, often multi-site	Largest **pre-approval** trial
Phase IV	Postmarketing surveillance after approval	The general prescribed population	**After approval**; catches rare effects

How to tell them apart. Sources: S13; S12 (notes)

Efficacy vs. effectiveness

	Efficacy	Effectiveness
Setting	Ideal, controlled trial conditions	Everyday clinical practice
What changes it	Not covered in materials	Real-world adherence, comorbidities (and drug interactions — SG)
Shorthand	The trial result	What happens with real patients

How to tell them apart. Sources: S14; S14 (notes); SG U1-R4

Major US drug laws (textbook Table 1–6)

Law	What it did
Food, Drug, and Cosmetic Act (1938)	New drugs must be safe (not yet proven effective); followed sulfanilamide deaths
Durham-Humphrey Act (1952)	FDA decides which drugs need a prescription
Kefauver-Harris Amendments (1962)	Proof of efficacy as well as safety; adverse-reaction reporting (after thalidomide)
Comprehensive Drug Abuse Prevention and Control Act (1970)	Drug schedules and controls for habit-forming drugs
Orphan Drug Amendment (1983)	Incentives for diseases with fewer than 200,000 US patients
Drug Price Competition and Patent Restoration Act (1984)	Abbreviated applications for generics; bioequivalence data required
Dietary Supplement Health and Education Act (1994)	Supplements treated as food; no full FDA drug review
Biologics Price Competition and Innovation Act (2009)	Abbreviated pathway for biosimilars

TB How to tell them apart. Sources: p. 16

SLIDE DISCREPANCY

- **Phase I population**
- Slides: Phase I: a small group of healthy volunteers. S13
- Textbook: Phase I: 20–100 healthy volunteers — but if significant toxicity is expected (cancer, AIDS therapy), volunteer **patients with the disease** are used instead. p. 17

SLIDE DISCREPANCY

- **Phase II size**
- Slides: Phase II: a few hundred patients with the target condition. S13

- Textbook: Phase II: a modest number of patients (100–200) with the target disease. p. 17

WHY IT HAPPENS

- Why are rare adverse effects found after approval? Trials are too small and too short to catch 1-in-10,000 events — so they show up in Phase IV, in the general prescribed population. S12 (notes); S13

MNEMONIC (MEMORY AID)

- Phases by question: **I** = Is it safe (healthy volunteers)? **II** = Is there an effect? **III** = Is it confirmed at scale? **IV** = Is it still safe in everyone? S13
- Effica**C**y = **C**ontrolled conditions. Effecti**V**eness = real-world **V**ariables (adherence, comorbidities). S14

HIGH-YIELD

- **Efficacy** = ideal trial conditions; **effectiveness** = everyday practice (adherence, comorbidities). S14
- Phase I healthy volunteers → II few hundred patients → III hundreds–thousands (confirm) → IV postmarketing. S13
- Rare adverse effects (1 in 10,000) are often first found in Phase IV. S12 (notes)
- Risk-benefit is always comparative — ask "compared to what?" S15; S15 (notes)

3 PART 3 · TOPIC

Pharmacodynamics: Targets & Receptors

SO3 IO1.1

Key definitions

- **Drug target:** the specific molecule a drug binds to produce its effect. S19
- **Receptor:** a protein target whose activation (or blockade) triggers a cellular response. S19
- **Signal transduction:** the chain of molecular events that converts receptor binding into a cellular effect. S19
- **Pharmacologic response:** the measurable downstream result — contraction, secretion, gene transcription, and so on. S19
- Full agonist — maximal receptor activation. S22; SG U1-M
- Partial agonist — submaximal activation, even at full receptor occupancy. S22; SG U1-M
- Inverse agonist — reduces baseline (constitutive) receptor activity. S22; SG U1-M
- Competitive antagonist — reversibly competes with agonist at the same site; overcome by more agonist. S22; SG U1-M
- Noncompetitive antagonist — binds elsewhere or irreversibly reduces the maximal effect. S22; SG U1-M
- Irreversible antagonist — permanently inactivates the receptor until new receptor is synthesized. S22; SG U1-M
- TB **Kd:** the drug concentration that fills half the receptors; a low Kd means high affinity. p. 23

How it works

- The pathway: drug → target → signal transduction → response. Every pharmacodynamics topic for the rest of the course maps onto this chain. S19 (notes)
- Receptor concept: Drug + Receptor → Drug-Receptor Complex → Cellular Response. S21
- This single chain underlies agonism, antagonism, potency, and efficacy — all describe different features of this interaction. S21
- Five major drug target classes: receptors, ion channels, enzymes, transporters, nucleic acids. S20; SG U1-R5
- Receptors: G-protein-coupled, ligand-gated ion channel, enzyme-linked, nuclear. S20
- Ion channels: directly gate the flow of ions across a membrane. S20
- Enzymes: drugs can inhibit or, less commonly, activate an enzyme's activity. S20
- Transporters & nucleic acids: carrier proteins; or DNA/RNA for some antimicrobials & gene therapies. S20
- Most drugs target receptors or enzymes — ion channels and transporters matter especially for cardiac and CNS drugs. S20 (notes)
- An inverse agonist actively reduces constitutive (baseline) receptor activity below normal; a true antagonist merely blocks the agonist without lowering baseline activity itself. S28
- TB **Spare receptors:** a maximal response can occur without occupying every

receptor (e.g., about 90% of cardiac β receptors can be blocked and maximal inotropic response still preserved). p. 23

- TB **Irreversible antagonist** effects last until new receptors are made — duration depends on receptor turnover, not drug elimination (phenoxybenzamine for pheochromocytoma). p. 25
- TB A partial agonist's lower maximal effect is **not** due to lower affinity — it can occupy all receptors and still produce less response. p. 26
- TB **Allosteric modulators** bind a different site from the agonist (e.g., benzodiazepines enhance GABA's effect). p. 25
- TB Agonist-induced **desensitization** (tachyphylaxis) and downregulation reduce response with continued exposure. p. 39
- TB Targets beyond receptors: **enzymes** (statins on HMG-CoA reductase), **transporters** (digitalis on Na^+/K^+-ATPase; antidepressants) and **structural proteins** (colchicine on tubulin). p. 22
- TB Five signaling mechanisms carry a drug's message into the cell: intracellular receptor, receptor tyrosine kinase, cytokine receptor (JAK-STAT), ion channel and G protein (compared in the table). p. 28-32

Drugs named as examples

- Albuterol — selective β2-adrenergic agonist (example of agonism). S26
- TB **Chemical antagonism** needs no receptor: positively charged protamine binds and neutralizes negatively charged heparin. p. 26
- TB **Physiologic antagonism:** two drugs act on different receptors with opposing effects (e.g., insulin opposes glucocorticoid-induced hyperglycemia). p. 27

CLINICAL

- Albuterol is a selective β2-adrenergic receptor agonist. S26

- Binding activates the β2 receptor → increases intracellular cAMP. S26
- Rising cAMP relaxes bronchial smooth muscle → bronchodilation. S26
- **This is the same receptor-to-response logic you will reuse for every agonist drug this term.** S26
- Albuterol bridges introductory pharmacodynamics into the pulmonary pharmacology seen later in the curriculum. S26 (notes)
- **TB** Competitive antagonism is surmountable in real life: exercise or stress raises catecholamines enough to overcome a fixed propranolol dose. p. 25
- **TB** Propranolol markedly slows the heart rate in pheochromocytoma (high catecholamines) but not the resting rate of a trained marathon runner. p. 39
- **TB** Glucocorticoids will not stop an acute asthma attack within minutes: gene-acting drugs need hours to make new proteins. p. 28
- **TB** Phosphodiesterase inhibitors prolong second messengers: milrinone (cAMP) and sildenafil (cGMP). p. 34; p. 35

TRAP

- Classic exam trap: a competitive antagonist shifts the dose-response curve **right** (same Emax, more drug needed); a noncompetitive/irreversible antagonist **lowers the Emax** instead. S22 (notes)
- A rightward shift with an unchanged plateau is the graphical signature of competitive antagonism — potency falls, but because the antagonism is surmountable, enough agonist can still reach the original Emax. S25
- Inverse agonist ≠ antagonist: only the inverse agonist lowers baseline activity below normal, without any agonist present. S27-28
- **SG** Study-guide distractor: "has no effect on receptor activity whatsoever" matches none of the agonist/antagonist types. SG U1-M

Step by step

The pharmacodynamic pathway S19 (notes); S21

Drug → Target (e.g., receptor) → Signal transduction → Pharmacologic response

Albuterol: receptor → second messenger → effect S26

Albuterol binds β2-adrenergic receptor → β2 receptor activated → ↑ intracellular cAMP → Bronchial smooth muscle relaxes → Bronchodilation

Competitive antagonism S22; S22 (notes); S25

Antagonist reversibly occupies the agonist's site →

More agonist needed for same effect → Curve shifts right →

Same Emax still reachable (surmountable)

Figure 1 · Agonist & antagonist effects. Original illustrative curves (not Katzung data): a competitive antagonist shifts the curve right with the same Emax; a noncompetitive/irreversible antagonist lowers Emax instead. Read left to right: agonist alone → competitive (shifted right, same ceiling) → noncompetitive (lower ceiling). S23; S23 (notes)

Agonists vs. antagonists

Type	What it does at the receptor	Dose-response curve	How to tell apart
Full agonist	Maximal receptor activation	Not covered in materials	Reaches maximal activation
Partial agonist	Submaximal activation	Not covered in materials	Submaximal **even at full occupancy**
Inverse agonist	Reduces baseline (constitutive) activity	Not covered in materials	Lowers activity **below normal with no agonist present**

Type	What it does at the receptor	Dose-response curve	How to tell apart
Competitive antagonist	Reversibly competes at the same site	**Shifts right, same Emax**	Overcome by more agonist (surmountable)
Noncompetitive antagonist	Binds elsewhere or irreversibly	**Lowers Emax**	Plateaus lower instead of shifting right
Irreversible antagonist	Permanently inactivates the receptor	**Lowers Emax**	Lasts until new receptor is synthesized

How to tell them apart. Sources: S22; S22 (notes); S25; S27-28

The five major drug target classes

Target	What it is / does	Note
Receptors	G-protein-coupled, ligand-gated ion channel, enzyme-linked, nuclear	Most common target (with enzymes)
Ion channels	Directly gate the flow of ions across a membrane	Especially cardiac and CNS drugs
Enzymes	Drugs inhibit or, less commonly, activate them	Most common target (with receptors)
Transporters	Carrier proteins	Especially cardiac and CNS drugs
Nucleic acids	DNA/RNA	Some antimicrobials & gene therapies

How to tell them apart. Sources: S20; S20 (notes)

Five ways a signal crosses the membrane (textbook Ch. 2)

Mechanism	Examples	How it signals	Clinical point
Intracellular receptor	Corticosteroids, sex steroids, vitamin D, thyroid hormone	Lipid-soluble ligand enters the cell; receptor changes gene transcription	Effects lag 30 min to hours and persist after the drug is gone
Receptor tyrosine kinase	Insulin, EGF, PDGF	Receptors dimerize and phosphorylate tyrosines	Blocked in cancer (trastuzumab, gefitinib)
Cytokine receptor	Growth hormone, erythropoietin, interferons	Separate JAK kinase activates STATs, which enter the nucleus	Same pattern as tyrosine kinases, kinase is a separate protein
Ion channel	Nicotinic ACh, GABA, glutamate; voltage-gated Ca^{2+} and Na^{+} channels	Opens or blocks ion flow, changing membrane potential	Fastest signal (milliseconds); verapamil, local anesthetics

Mechanism	Examples	How it signals	Clinical point
G protein–coupled receptor	β and α adrenoceptors, muscarinic ACh, opioids	G protein (GTP) changes an effector, which changes a second messenger	Largest family; desensitizes with continued agonist

TB How to tell them apart. Sources: p. 28-32

G proteins and their second messengers (textbook Table 2–1)

G protein	Receptors for	Effect
Gs	β-adrenergic amines, histamine, glucagon	↑ adenylyl cyclase → ↑ cAMP
Gi	α_2-adrenergic amines, muscarinic ACh, opioids	↓ cAMP; opens cardiac K^+ channels → ↓ heart rate
Gq	Muscarinic ACh, serotonin (5-HT_2)	↑ phospholipase C → IP_3, DAG, ↑ cytoplasmic Ca^{2+}

TB How to tell them apart. Sources: p. 32

WHY IT HAPPENS

- Why can more agonist overcome a competitive antagonist? Both compete **reversibly for the same site** — flood the site with agonist and it wins, so the curve shifts right but the same Emax is still reached. S22; S25
- Why does albuterol open the airways? β2 activation → ↑ intracellular cAMP → bronchial smooth muscle relaxes → bronchodilation. S26

MNEMONIC (MEMORY AID)

- **C**ompetitive = **C**an be outcompeted (shift right). **N**oncompetitive = **N**o full ceiling (lower Emax). S22 (notes)

HIGH-YIELD

- Drug → target → signal transduction → response. S19 (notes)
- Five target classes: receptors, ion channels, enzymes, transporters, nucleic acids. S20; SG U1-R5
- Competitive antagonist = curve **shifts right, same Emax**; noncompetitive/irreversible = **lower Emax**. S22 (notes)
- Inverse agonist lowers baseline activity below normal; an antagonist only blocks. S28
- Pharmacodynamics explains how drugs bind receptors to produce agonism or antagonism. S56

4 PART 4 · TOPIC

Dose-Response Relationships

SO4 IO1.1

Key definitions

- **Potency:** how much drug is required to produce a given effect — a more potent drug needs a lower dose. S30; SG U1-R6
- **Efficacy (Emax):** the maximum effect a drug can produce, no matter how high the dose goes. S30; SG U1-R6
- **EC_{50}:** the drug concentration producing 50% of the maximal effect — the standard measure of potency. S30
- **Therapeutic Index (TI) = TD_{50} ÷ ED_{50} (toxic dose in 50% ÷ effective dose in 50%).** S38; SG U1-R7
- TB **Quantal dose-response:** ED_{50} (effective in 50%), TD_{50} (toxic in 50%), LD_{50} (lethal in 50% of animals). p. 38
- TB **Therapeutic window:** the range between the minimum therapeutic dose and the minimum toxic dose — more practical than TI for dosing a patient. p. 38

How it works

- The sigmoidal dose-response curve: x-axis = log(dose), y-axis = effect. EC_{50} is on the x-axis; Emax is the plateau — the ceiling effect. S30 (notes); S31
- EC_{50} is read off the curve at the halfway point between baseline and Emax. S33
- Lower EC_{50} = more potent. Higher Emax = more efficacious. S34
- Potency and efficacy are independent properties — a drug can differ in one without differing in the other. S35
- A curve farther left reaches the same effect at a lower dose (greater potency); curves reaching the same plateau height have identical Emax (efficacy). S37
- A wide TI means a large margin between an effective dose and a toxic dose — generally safer to dose. S38
- A narrow TI means effective and toxic doses are close together — requires careful dosing and often blood-level monitoring. S38; SG U1-R7
- The distance between the ED_{50} and TD_{50} markers is the therapeutic index itself. S41
- Dose-response curves describe potency and efficacy. S56

Drugs named as examples

- **Classic narrow-TI drugs: warfarin, lithium, digoxin, phenytoin.** S38; SG U1-R7

CLINICAL

- Potency clinically matters for dosing — not for choosing the "better" drug. S34

- Efficacy clinically matters much more than potency — a highly potent drug with low efficacy may be clinically useless. S34
- With a narrow TI, a small dosing error can move a patient from effective to toxic — which is why these drugs require blood-level monitoring. S41; S39 (notes)
- Students should be able to name all four narrow-TI drugs without hesitation and know that each requires therapeutic drug monitoring. S38 (notes)
- TB No drug causes only a single specific effect — drugs are **selective**, not specific. p. 40
- TB Asthma + hypertension on propranolol: nonselective β blockade can cause bronchoconstriction; a β_1-selective blocker (metoprolol) is an alternative. p. 41

TRAP

- A drug can be very potent (tiny dose) but have low efficacy — potency alone never tells you whether a drug "works better." S34 (notes)
- Same Emax + lower dose needed = MORE POTENT, EQUALLY efficacious — not "more efficacious." S43; SG U1-R6
- EC_{50} measures potency — not toxicity and not bioavailability. S33
- A narrow-TI drug is not "safer because it needs a lower dose" — it has a smaller margin and needs closer monitoring. S40-41

Step by step

TB Asthma + propranolol (textbook case) p. 41

Nonselective β blocker (propranolol) → Blocks β2 in the airways →

Bronchoconstriction / no bronchodilation →

Use a β1-selective blocker (metoprolol) or another antihypertensive

Figure 2 · The sigmoidal dose-response curve. Original illustrative curve (illustrative values, not from any textbook): EC_{50} marks 50% of Emax; Emax is the plateau — the ceiling effect. S31; S31 (notes)

Figure 3 · Potency vs. efficacy, side by side. Left panel: two drugs with the same Emax but different EC_{50} (potency differs). Right panel: two drugs with the same EC_{50} but different Emax (efficacy differs). Potency and efficacy are independent properties. S35; S35 (notes)

Figure 4 · Wide vs. narrow therapeutic index. Original illustrative number-line comparison: a wide gap between ED_{50} and TD_{50} is safer to dose than a narrow one (illustrative values). Note how close ED_{50} and TD_{50} sit for the narrow-TI example (e.g., warfarin, lithium, digoxin, phenytoin). S39; S39 (notes)

TB **Textbook concept · Quantal dose-response and the therapeutic window.** Original diagram (illustrative values) of a textbook concept: the % of a population showing the therapeutic effect (blue) or a toxic effect (red) as dose rises. ED_{50} and TD_{50} give TI = TD_{50} ÷ ED_{50} (an animal-study measure); the therapeutic window — minimum therapeutic dose to minimum toxic dose — is the more practical range for dosing a patient. p. 38

Potency vs. efficacy

	Potency	Efficacy (Emax)
Definition	How much drug is needed to get an effect	The maximum effect the drug can produce
Measured by	EC_{50} — **lower EC_{50} = more potent**	Plateau height — **higher Emax = more efficacious**
On the curve	Curve farther **left**	Curve plateaus **higher**
Clinical weight	Matters for **dosing**, not for picking the "better" drug	Matters **much more** — potent + low efficacy may be useless

How to tell them apart. Sources: S30; S34; S37

Wide vs. narrow therapeutic index

	Wide TI	Narrow TI
ED_{50} vs. TD_{50}	Far apart — large safety margin	Close together — small safety margin
Dosing	Generally safer to dose	Careful dosing; small errors → toxic
Monitoring	Not covered in materials	Often **blood-level monitoring**

	Wide TI	Narrow TI
Examples	Not covered in materials	**Warfarin, lithium, digoxin, phenytoin**

How to tell them apart. Sources: S38; S39; S41

The narrow-TI drugs in numbers (textbook Table 3–1)

Drug	Half-life	Target level	Toxic above	Worth knowing
Digoxin	39 h	1 ng/mL	2 ng/mL	Vd ~500 L (sequestered in tissue); 67% excreted in urine
Lithium	22 h	0.7 mEq/L	2 mEq/L	95% excreted in urine, 0% protein-bound — renal function rules
Phenytoin	Concentration-dependent	10 mg/L	20 mg/L	Saturable (capacity-limited) elimination; 89% protein-bound
Warfarin	37 h	Not in table	Not in table	99% protein-bound

TB How to tell them apart. Sources: p. 44; p. 45

WHY IT HAPPENS

- Why do narrow-TI drugs need blood levels? The effective dose and the toxic dose sit so close together that a small dosing error can push the patient from effective to toxic. S38; S41
- Why does efficacy beat potency clinically? Potency only changes the dose you give; efficacy sets the ceiling of what the drug can achieve at any dose. S30; S34

MNEMONIC (MEMORY AID)

- Narrow-TI drugs — "**W**atch **L**evels **D**uring **P**rescribing": **W**arfarin, **L**ithium, **D**igoxin, **P**henytoin. S38
- TI = **T**oxic over **E**ffective → TD_{50} ÷ ED_{50} (toxic on top). S38

HIGH-YIELD

- **Potency** = dose needed (EC_{50}, lower = more potent); **efficacy** = Emax (ceiling). S30; S34
- Potency and efficacy are independent; efficacy matters more clinically. S34; S35
- **TI = TD_{50} ÷ ED_{50}**; narrow TI → careful dosing + blood-level monitoring. S38
- Narrow-TI drugs: **warfarin, lithium, digoxin, phenytoin**. S38

5 PART 5 · TOPIC

Pharmacokinetics: ADME — Absorption & Distribution

SO5 IO1.2 IO1.3 IO1.5

Key definitions

- A — Absorption: movement of drug from the site of administration into the bloodstream. S45
- D — Distribution: movement of drug from blood into tissues. S45
- M — Metabolism: chemical transformation of drug, mainly in the liver. S45
- E — Excretion: removal of drug/metabolites from the body, mainly via the kidney. S45
- **Bioavailability (F) = the fraction of an administered dose that reaches systemic circulation unchanged.** S50; SG U1-R8
- First-pass effect: the liver metabolizes (can inactivate a large fraction of) an orally absorbed drug before it reaches systemic circulation. S51; S51 (notes); SG U1-R9
- **Volume of distribution (Vd):** a calculated value relating total drug in the body to plasma concentration — a high Vd suggests extensive tissue distribution. S52
- TB **pKa:** the pH at which a drug is half ionized and half nonionized. p. 10

How it works

- ADME is the organizing skeleton for the rest of the lecture — each remaining pharmacokinetics section is one ADME letter in depth. S45 (notes)
- ADME determines how much drug reaches its target and for how long. S56
- Metabolism (M arrow → liver) and Excretion (E arrow → kidney) are the back half of ADME — how the body gets rid of the drug; Absorption and Distribution govern how the drug arrives and spreads. S46; S48
- First-pass route: oral drug → GI tract → portal circulation → liver → systemic circulation. S51
- By definition, an IV dose has 100% bioavailability — the entire dose enters the bloodstream directly (no absorption step). S50; SG U1-R8
- Oral bioavailability is reduced by incomplete absorption and by first-pass hepatic metabolism. S50
- **Plasma protein binding:** only unbound ("free") drug is pharmacologically active and able to cross membranes. S52
- **Lipid solubility:** lipophilic drugs cross membranes more readily and distribute into fatty tissue. S52
- **Tissue perfusion:** highly perfused organs (brain, heart, kidney) receive drug faster than poorly perfused tissue (fat, bone). S52
- Vd is a theoretical volume, not a real anatomic space. S52 (notes)
- Blood-brain barrier — **Why can some drugs readily enter the CNS while others**

cannot? Tight junctions between endothelial cells restrict paracellular movement. S53

- Small, lipophilic, non-ionized drugs cross the BBB more easily than large, polar, or highly protein-bound drugs. S53
- Efflux transporters (like P-glycoprotein) can actively pump some drugs back out of the CNS. S53
- TB First-pass loss can happen in the **gut wall** (e.g., intestinal CYP3A4) as well as the liver; grapefruit juice inhibits gut-wall metabolism and P-glycoprotein, raising absorption of some drugs. p. 48
- TB **P-glycoprotein (MDR1)** pumps drugs out of cells in brain, testes, and gut; inhibiting it raises digoxin blood levels. p. 9
- TB **Ion trapping:** weak acids are excreted faster in **alkaline** urine; weak bases faster in **acidic** urine. p. 10; p. 11
- TB Drugs cross membranes by **lipid diffusion** (the main route), aqueous pores, **carriers** (saturable) or endocytosis (vitamin B_{12}, iron). p. 8; p. 9
- TB Only the **nonionized** form crosses lipid membranes: a weak acid is mostly nonionized when pH < pKa, a weak base when pH > pKa (Henderson-Hasselbalch). p. 9; p. 10

Drugs named as examples

- Large first-pass effect: nitroglycerin, propranolol, morphine — all require higher oral doses (or a non-oral route) to achieve the same systemic exposure as IV. S51; SG U1-R9
- Very high Vd: digoxin — mostly sequestered in tissue, not floating around in plasma. S52 (notes)
- Subcutaneous route is common for insulin and heparin. S49
- P-glycoprotein — an efflux transporter at the BBB. S53
- TB Weak acids: aspirin (pKa 3.5), warfarin (5.0). Weak bases: morphine (7.9), propranolol (9.4). p. 10

CLINICAL

- Routes that bypass first-pass metabolism: essentially everything except oral — IV, sublingual, inhaled, transdermal, topical, SC, IM. S49 (notes)
- Sublingual absorption enters systemic venous circulation directly, bypassing the portal vein and the liver's first-pass metabolism; oral dosing passes through the liver first. S55; SG U1-R9
- Because the liver can inactivate a large fraction of drug first, some drugs are given sublingually or transdermally instead of orally. S51 (notes)
- A highly polar antibiotic would generally not cross the BBB well — unless the meninges are inflamed, as in meningitis. S53 (notes)
- **TB** Aspirin overdose: sodium bicarbonate alkalinizes blood and urine, reducing aspirin entry into tissues and speeding its renal excretion. p. 20

TRAP

- IV = 100% bioavailable with no absorption phase; oral is less than 100%. S49; S50
- Taking an oral drug with food or on an empty stomach does not bypass first-pass metabolism — sublingual does. S54-55
- A very high Vd does not mean a big anatomic space — it means the drug is mostly in tissue, not plasma. S52 (notes)
- Only free (unbound) drug is active and can cross membranes — protein-bound drug cannot. S52

Step by step

ADME S45; S46

Dose → Absorption → blood → Distribution → tissues → Metabolism (mainly liver) → Excretion (mainly kidney)

First-pass effect (oral route) S51

Oral drug → GI tract → Portal circulation → Liver (first-pass metabolism) → Systemic circulation

TB **Aspirin overdose and urine pH (textbook case)** p. 20

Aspirin = weak acid → Acidosis drives it into tissues and back from urine →

Sodium bicarbonate alkalinizes blood and urine →

Drug ionized → trapped in urine → faster renal clearance

Figure 5 · The ADME compartment model. Original illustrative compartment diagram: how a dose moves from administration through distribution, metabolism, and excretion. Dose → Absorption → Central compartment (blood/plasma) ⇄ Peripheral compartment (tissues); M arrow → Metabolism (liver), E arrow → Excretion (kidney). S46; S46 (notes)

Routes of administration

Route	Key feature	Bypasses first-pass?
Oral (PO)	Convenient; subject to first-pass metabolism and variable absorption	**No**
Sublingual	Absorbed directly into systemic circulation	Yes
Intravenous (IV)	100% bioavailable; immediate onset; no absorption phase	Yes
Intramuscular (IM)	Faster than oral; good for depot/long-acting formulations	Yes
Subcutaneous (SC)	Slower, steady absorption; common for insulin, heparin	Yes

Route	Key feature	Bypasses first-pass?
Inhaled	Rapid local effect in the lungs; minimizes systemic exposure for some drugs	Yes
Transdermal	Slow, sustained systemic absorption through the skin	Yes
Topical	Local effect at the site of application; minimal systemic absorption	Yes

How to tell them apart. Sources: S49; S49 (notes)

Who crosses the blood-brain barrier?

	Crosses more easily	Crosses less easily
Size	Small	Large
Chemistry	Lipophilic, non-ionized	Polar
Protein binding	Not covered in materials	Highly protein-bound
Other	Not covered in materials	Pumped out by efflux transporters (P-glycoprotein)

How to tell them apart. Sources: S53

Bioavailability by route (textbook Table 3–3)

Route	Bioavailability (%)	Note
Intravenous (IV)	100	Most rapid onset
Intramuscular (IM)	75 to ≤100	Large volumes often feasible; may be painful
Subcutaneous (SC)	75 to ≤100	Smaller volumes than IM; may be painful
Oral (PO)	5 to <100	Most convenient; first-pass effect may be important
Rectal (PR)	30 to <100	Less first-pass effect than oral
Inhalation	5 to <100	Often very rapid onset
Transdermal	80 to ≤100	Usually very slow absorption; prolonged duration

TB How to tell them apart. Sources: p. 48

WHY IT HAPPENS

- Why is IV bioavailability 100%? The whole dose enters the bloodstream directly — there is no absorption step and no trip through the liver first. S50; S49
- Why does sublingual skip first-pass? It drains into systemic veins, not the portal vein — so the drug reaches the body before the liver can inactivate it. S55; S51

- Why do highly protein-bound drugs act less and cross the BBB poorly? Only free (unbound) drug is active and able to cross membranes. S52; S53

MNEMONIC (MEMORY AID)

- Big first-pass drugs "**N**eed **P**lenty **M**ore by mouth": **N**itroglycerin, **P**ropranolol, **M**orphine. S51

HIGH-YIELD

- **F** = fraction of dose reaching systemic circulation unchanged; IV = 100%. S50
- First-pass: oral → GI → portal → liver → systemic; big examples nitroglycerin, propranolol, morphine. S51
- Every route except oral bypasses first-pass (sublingual = direct to systemic veins). S49 (notes); S55
- Only **free (unbound)** drug is active; high **Vd** = drug mostly in tissue (e.g., digoxin). S52; S52 (notes)
- BBB: small, lipophilic, non-ionized drugs cross; P-glycoprotein pumps some back out. S53

Cram sheet

Every term, mechanism, drug, and number. Hide the answers and quiz yourself; tap a blurred answer to peek.

Prompt	Answer
Pharmacology	Study of how drugs interact with living systems — molecule to whole patient S5
Drug (medication)	Chemical used to diagnose, treat, cure, prevent disease, or alter a physiologic process S5
Pharmacologic vs. therapeutic effect	Any measurable biological change vs. the intended, beneficial effect S5
Adverse drug reaction (ADR)	Unintended, harmful response at **normal therapeutic doses** S5
Toxicity	Harm when dose/exposure **exceeds the therapeutic range** S5
Chemical / generic / brand name	Exact structure (rarely used) / official non-proprietary, one per drug / manufacturer's, many, vary by country S7
Why prescribe by generic name	Unambiguous, international; avoids hidden duplicate therapy (double-dosing) S7; S9
Tylenol	Brand name of acetaminophen S7
Pharmacokinetics vs. pharmacodynamics	Body → drug (ADME) vs. drug → body (receptors & effect) S6
Pharmacogenomics	How genetic variation shapes individual drug response S6
Clinical pharmacology / toxicology	Applying PK/PD to real patients; the study of poisoning S6
4 sources of new drugs	Natural products · synthetic & biotech · biologics · drug repurposing S11
Biologics	Monoclonal antibodies, gene- and cell-based therapies S11
Drug pipeline	Discovery → preclinical → clinical trials → FDA review → approval → postmarket surveillance S12
Phase I	Safety, tolerability, dose-finding — small group of **healthy volunteers** S13
Phase II	Preliminary efficacy & side effects — a few hundred patients with the condition S13
Phase III	Confirm efficacy, monitor AEs at scale — hundreds to thousands, multi-site S13
Phase IV	Postmarketing surveillance — general prescribed population S13

Prompt	Answer
Rare AE (1 in 10,000)	Often first found in Phase IV — trials too small/short S12 (notes)
Efficacy vs. effectiveness	Ideal trial conditions vs. everyday practice (adherence, comorbidities) S14
FDA MedWatch	Adverse-event reporting system (voluntary + mandatory) S14
Risk-benefit case: 70% improve, 1% serious hepatotoxicity	Always comparative — "compared to what?" S15; S15 (notes)
5 drug target classes	Receptors · ion channels · enzymes · transporters · nucleic acids S20
Receptor types	G-protein-coupled · ligand-gated ion channel · enzyme-linked · nuclear S20
Most common targets	Receptors and enzymes (ion channels/transporters: cardiac & CNS drugs) S20 (notes)
Full / partial / inverse agonist	Maximal activation / submaximal even at full occupancy / lowers baseline activity S22
Competitive antagonist	Same site, reversible, overcome by more agonist → **curve right, same Emax** S22; S22 (notes)
Noncompetitive / irreversible antagonist	Binds elsewhere or irreversibly → **lower Emax**; irreversible lasts until new receptor made S22; S22 (notes)
Albuterol	Selective β2 agonist → ↑ cAMP → bronchial smooth muscle relaxes → bronchodilation S26
Potency	Dose needed for an effect — **lower EC_{50} = more potent** S30; S34
Efficacy (Emax)	Maximum possible effect — **higher Emax = more efficacious**; matters more clinically S30; S34
EC_{50}	Concentration giving 50% of Emax — standard measure of potency S30
Therapeutic index	**TI = TD_{50} ÷ ED_{50}**; narrow = careful dosing + blood levels S38
Narrow-TI drugs	**Warfarin, lithium, digoxin, phenytoin** S38
ADME	Absorption → distribution → metabolism (mainly liver) → excretion (mainly kidney) S45
Bioavailability (F)	Fraction of dose reaching systemic circulation unchanged; **IV = 100%** S50
Why oral F < 100%	Incomplete absorption + first-pass hepatic metabolism S50
First-pass path	Oral → GI tract → portal circulation → liver → systemic circulation S51
Big first-pass drugs	Nitroglycerin, propranolol, morphine — need higher oral doses S51

Prompt	Answer
Routes that bypass first-pass	Every route except oral (IV, sublingual, inhaled, transdermal, topical, SC, IM) S49 (notes)
Sublingual	Directly into systemic venous circulation — skips portal vein & liver S49; S55
IM / SC / transdermal	Faster than oral, depot · slower steady (insulin, heparin) · slow sustained S49
Inhaled / topical	Rapid local lung effect · local effect, minimal systemic absorption S49
Plasma protein binding	Only **free (unbound)** drug is active and crosses membranes S52
Tissue perfusion	Brain, heart, kidney get drug fast; fat, bone slowly S52
Volume of distribution (Vd)	Total drug in body vs. plasma concentration; theoretical; high = in tissue (digoxin) S52; S52 (notes)
Blood-brain barrier	Tight junctions; small, lipophilic, non-ionized cross; P-glycoprotein pumps out S53
Polar antibiotic + meningitis	Usually poor BBB entry — better when meninges are inflamed S53 (notes)
Phase I size (textbook)	20–100 healthy volunteers; patients if drug is very toxic (cancer, AIDS) p. 17
Phase with highest failure	Phase II — only ~25% move to Phase III p. 17
Kefauver-Harris (1962)	Required proof of **efficacy** (thalidomide) p. 16
Generic approval	ANDA with bioequivalence data; patent life 20 years p. 16; p. 18
Spare receptors	Max response without full occupancy (~90% cardiac β can be blocked) p. 23
Chemical antagonist	Protamine neutralizes heparin — no receptor p. 26
Therapeutic window	Min therapeutic → min toxic dose; more practical than TI p. 38
Aspirin overdose	Sodium bicarbonate → alkaline urine → faster excretion p. 20
Weak bonds vs covalent	Weak = more selective; covalent = long-lasting (aspirin) p. 4
pKa	pH where ionized = nonionized p. 10
Ion trapping	Acids leave faster in alkaline urine; bases in acidic urine p. 10
Intracellular receptors	Steroids, thyroid, vitamin D; lag 30 min–hours, effect persists p. 28
Gs / Gi / Gq	↑ cAMP / ↓ cAMP / ↑ IP_3, DAG, Ca^{2+} p. 32
Gold-standard trial	Randomized controlled trial p. 15

Self-test

First, the lecture's own 10 knowledge checks — exactly as written. Then original PANCE-style questions. Tap an answer to reveal the explanation.

From the lecture

Lecture check 1 · S8-9

A patient says they take no medications, but separately mentions using "Tylenol" for headaches and a combination cold-and-flu product. What does this scenario illustrate?

- **A** Brand names can obscure a shared generic ingredient, risking unintentional double-dosing
- **B** Chemical names are standardized internationally, so this mix-up could not happen
- **C** Generic names vary by manufacturer, which caused the confusion
- **D** Pharmacologic effect and therapeutic effect are simply two words for the same thing

ANSWER: A

Different brand-name products (Tylenol, many combination cold/flu products) can share the same active generic ingredient. Prescribing and counseling by generic name avoids hiding duplicate therapy — the reason PAs default to generic names. S8-9

Lecture check 2 · S16-17

In which clinical trial phase is a drug first tested in a large patient population specifically to confirm efficacy and monitor for adverse effects at scale, just before approval?

- **A** Phase I
- **B** Phase II
- **C** Phase III
- **D** Phase IV

ANSWER: C

Phase III trials enroll hundreds to thousands of patients, often at multiple sites, to confirm efficacy and monitor safety at scale before approval. Phase I is small-scale safety/dosing, Phase II is preliminary efficacy, and Phase IV is postmarketing surveillance after approval. S16-17

Lecture check 3 · S24-25

In Figure 1 above, the dashed curve (+ Competitive antagonist) is shifted to the right of the agonist-alone curve but reaches the same plateau at the top. What does this pattern show?

- **A** More agonist is needed to reach the same effect, but the same maximal effect is still achievable
- **B** The competitive antagonist lowers the drug's efficacy (Emax)
- **C** The competitive antagonist increases the drug's potency
- **D** The competitive antagonist has no measurable effect on the agonist's curve

ANSWER: A

A rightward shift with an unchanged plateau is the graphical signature of competitive antagonism: it takes more drug to reach a given effect (potency falls), but because the antagonism is surmountable, enough agonist can still reach the original Emax. Contrast this with the noncompetitive/irreversible curve in the same figure, which plateaus lower instead of shifting right. S24-25

Lecture check 4 · S27-28

A drug binds the same receptor as an endogenous agonist and lowers the tissue's baseline (constitutive) activity below normal, without any agonist present. This drug is best classified as a(n):

- **A** Full agonist
- **B** Partial agonist
- **C** Competitive antagonist
- **D** Inverse agonist

ANSWER: D

An inverse agonist actively reduces constitutive (baseline) receptor activity below normal — a true antagonist merely blocks the agonist without lowering baseline activity itself. S27-28

Lecture check 5 · S32-33

On the dose-response curve in Figure 2, the point marked EC_{50} sits exactly halfway up the rising part of the curve. What does this point represent?

- **A** The dose at which the drug becomes toxic
- **B** The concentration producing 50% of the drug's maximal effect
- **C** The point where the drug has zero effect
- **D** The drug's bioavailability

ANSWER: B

EC_{50} is read directly off the curve at the halfway point between baseline and Emax (the plateau) — it is the standard graphical measure of potency, not a measure of toxicity or bioavailability. S32-33

Lecture check 6 · S36-37

In the left panel of Figure 3, Drug A's curve sits to the left of Drug B's curve, but both curves plateau at the same height. What does this tell you?

- **A** Drug A is more potent than Drug B, and both have equal efficacy
- **B** Drug A is more efficacious than Drug B
- **C** Drug A and Drug B have different Emax values
- **D** Drug B is more potent than Drug A

ANSWER: A

A curve positioned farther left reaches the same effect at a lower dose — that is greater potency. Because both curves reach the same plateau height, their Emax (efficacy) is identical; only potency differs, exactly as the right panel then shows the reverse case (same potency, different efficacy). S36-37

Lecture check 7 · S40-41

Comparing the two number lines in Figure 4, the narrow-TI drug's ED_{50} and TD_{50} markers sit much closer together than the wide-TI drug's. What is the clinical implication?

- **A** The narrow-TI drug is safer because it requires a lower dose
- **B** The narrow-TI drug has a smaller margin between an effective and a toxic dose, requiring closer monitoring
- **C** The wide-TI drug is more likely to cause toxicity
- **D** Therapeutic index has no bearing on dosing safety

ANSWER: B

The distance between the ED_{50} and TD_{50} markers on the number line is the therapeutic index itself — a narrow gap (as with warfarin, lithium, digoxin, phenytoin) means a small dosing error can move a patient from effective to toxic, which is why these drugs require blood-level monitoring. S40-41

Lecture check 8 · S42-43

Drug X requires a much lower dose than Drug Y to achieve the same effect, but both drugs achieve the same maximum effect (Emax). Which statement is correct?

- **A** Drug X is more efficacious than Drug Y
- **B** Drug X is more potent than Drug Y
- **C** Drug X has a narrower therapeutic index than Drug Y
- **D** Drug X and Drug Y have different Emax values

ANSWER: B

Needing a lower dose for the same effect is the definition of greater potency. Since both drugs reach the same Emax, their efficacy is equal — potency and efficacy are independent properties. S42-43

Lecture check 9 · S47-48

In Figure 5, two separate arrows leave the Central Compartment box: one labeled "M" pointing to Metabolism (Liver), and one labeled "E" pointing to Excretion (Kidney). Together, these two arrows represent which part of ADME?

- **A** Absorption and Distribution
- **B** Distribution only
- **C** Metabolism and Excretion — how the drug is transformed and cleared
- **D** Only the Excretion step

ANSWER: C

The diagram shows drug leaving the Central Compartment by two separate routes: the "M" arrow to the liver (biotransformation) and the "E" arrow to the kidney (elimination). Together these are the back half of ADME — how the body gets rid of the drug, as opposed to Absorption and Distribution, which govern how the drug arrives and spreads. S47-48

Lecture check 10 · S54-55

Which route of administration bypasses first-pass hepatic metabolism entirely, by entering systemic venous circulation directly?

- **A** Sublingual
- **B** Oral, taken on an empty stomach
- **C** All routes undergo first-pass metabolism equally
- **D** Oral, taken with food

ANSWER: A

Sublingual absorption enters systemic venous circulation directly, bypassing the portal vein and the liver's first-pass metabolism — unlike oral dosing, which is absorbed into the portal circulation and passes through the liver first. S54-55

Original questions

Q1 · Clinical vignette · The Language of Pharmacology

A 45-year-old patient takes a new medication exactly as prescribed. Drug levels are within the therapeutic range, yet the patient develops an unintended, harmful response. Which term best describes this event?

- **A** Toxicity, because any harmful drug response counts as toxicity
- **B** An adverse drug reaction, because it occurred at a normal therapeutic dose
- **C** A therapeutic effect, because the drug was taken exactly as prescribed
- **D** A pharmacologic effect only, because it was a measurable biological change
- **E** Irreversible antagonism, because the receptor was permanently inactivated

ANSWER: B

An ADR is an unintended and harmful response occurring at normal therapeutic doses (S5). S5

- **A.** Toxicity is harm that occurs as dose or exposure exceeds the therapeutic range (S5); here levels were within range.
- **C.** A therapeutic effect is the intended, beneficial effect (S5); this response was unintended and harmful.
- **D.** Every ADR is a pharmacologic effect, but 'unintended and harmful at normal doses' is the more specific, best answer (S5).
- **E.** Irreversible antagonism describes receptor inactivation until new receptor is synthesized (S22) — not a type of adverse event.

Q2 · Mechanism · The Language of Pharmacology

Which statement correctly distinguishes pharmacokinetics from pharmacodynamics?

A Pharmacokinetics studies receptor binding; pharmacodynamics studies absorption and excretion

B Pharmacokinetics applies only to IV drugs; pharmacodynamics applies only to oral drugs

C Pharmacokinetics studies genetic variation; pharmacodynamics studies poisoning and overdose

D Pharmacokinetics is what the drug does to the body; pharmacodynamics is what the body does to the drug

E Pharmacokinetics is what the body does to the drug; pharmacodynamics is what the drug does to the body

ANSWER: E

PK = what the body does to the drug (ADME); PD = what the drug does to the body (receptors & effect) (S6). S6

- **A.** Receptors & effect belong to PD; absorption and excretion (ADME) belong to PK (S6).
- **B.** Neither branch is defined by route; PK covers ADME for any route (S6, S49).
- **C.** Genetic variation = pharmacogenomics; poisoning = clinical pharmacology/toxicology (S6).
- **D.** This reverses the definitions (S6).

Q3 · Clinical vignette · Drug Development & Regulation

A drug relieved symptoms in most participants of a controlled multi-site trial. In your clinic, far fewer patients improve; many miss doses and have several comorbidities. Which concept best explains the gap?

- **A** A narrow therapeutic index that appeared after approval
- **B** Lower potency than was measured during trial dose-finding
- **C** Extensive first-pass metabolism in the clinic's patients
- **D** Competitive antagonism of the drug by the comorbid conditions
- **E** Real-world effectiveness falling below the drug's trial efficacy

ANSWER: E

Efficacy is performance under ideal trial conditions; effectiveness is performance in everyday practice with real-world adherence and comorbidities (S14). S14; S14 (notes)

- **A.** TI describes the gap between effective and toxic doses (S38), not missed doses or comorbidities.
- **B.** Potency is the dose needed for an effect and matters for dosing, not for whether patients improve (S34).
- **C.** First-pass effect lowers oral bioavailability (S50-51); it doesn't explain adherence- and comorbidity-driven differences.
- **D.** Competitive antagonism is a drug competing at the same receptor site (S22), not a disease effect.

Q4 · Clinical vignette · Drug Development & Regulation

Two years after approval, reports reveal a serious reaction occurring in about 1 in 10,000 patients — never seen in the clinical trials. Which best explains why it was missed before approval?

- **A** Adverse-event reporting systems accept only voluntary reports
- **B** Approval came before the drug completed any clinical trial phase
- **C** Phase I trials enrolled patients who already had the target condition
- **D** Trial populations were too small and too short to detect so rare an event
- **E** Phase III trials are not designed to monitor adverse effects

ANSWER: D

Rare adverse effects are often first detected in Phase IV because trial populations are too small and too short in duration to catch 1-in-10,000 events (S12 notes). S12 (notes); S13; S14

- **A.** Reporting systems are both voluntary and mandatory (S14).
- **B.** Approved drugs pass through a structured pipeline with clinical trials before FDA review (S12, S56).
- **C.** Phase I uses a small group of healthy volunteers (S13).
- **E.** Phase III confirms efficacy AND monitors adverse effects at scale (S13).

Q5 · Mechanism · Drug Development & Regulation

According to the textbook, which clinical trial phase has the highest rate of drug failures?

- **A** Phase II
- **B** Phase IV
- **C** FDA review
- **D** Phase III
- **E** Phase I

ANSWER: A

Phase II has the highest failure rate; only about 25% of drugs move on to Phase III (p. 17). p. 17

- **B.** Phase IV is postmarketing surveillance after approval (p. 17).
- **C.** FDA review is the approval step after Phase III, not a trial phase (p. 17).
- **D.** Phase III confirms efficacy in thousands, after most failures have occurred (p. 17).
- **E.** Phase I sets the safe dose range in 20–100 subjects; it is not the main failure point (p. 17).

Q6 · Clinical vignette · Pharmacodynamics: Targets & Receptors

A 22-year-old patient inhales albuterol, and within minutes the bronchial smooth muscle relaxes. Which sequence best explains this effect?

A	β2 receptor blockade → falling intracellular cAMP → smooth muscle relaxation
B	Enzyme inhibition → falling intracellular cAMP → bronchial smooth muscle dilation
C	β2 receptor activation → rising intracellular cAMP → smooth muscle relaxation
D	Ion channel opening → rising intracellular cAMP → smooth muscle contraction
E	Inverse agonism at β2 → lowered baseline activity → smooth muscle relaxation

ANSWER: C

Albuterol is a selective β2 agonist: binding activates β2 → increases cAMP → relaxes bronchial smooth muscle → bronchodilation (S26). S26

- **A.** Albuterol is an agonist — it activates, not blocks, β2, and cAMP rises (S26).
- **B.** The mechanism is receptor activation with rising cAMP (S26).
- **D.** Albuterol acts on the β2 receptor, and the result is relaxation, not contraction (S26).
- **E.** Inverse agonists reduce baseline activity (S22); albuterol is a selective β2 agonist (S26).

Q7 · Mechanism · Pharmacodynamics: Targets & Receptors

A patient receives a drug that blocks a receptor. Raising the dose of the agonist medication eventually restores the full original maximal response. The blocking drug is most likely a(n):

A Noncompetitive antagonist

B Inverse agonist

C Irreversible antagonist

D Competitive antagonist

E Partial agonist

ANSWER: D

A competitive antagonist reversibly competes at the same site and is overcome by more agonist — same Emax, curve shifted right (S22, S22 notes). S22; S22 (notes)

- **A.** Noncompetitive antagonists reduce the maximal effect, so full Emax is not restored (S22).
- **B.** An inverse agonist lowers baseline activity; the scenario describes surmountable blockade (S22, S28).
- **C.** Irreversible antagonists permanently inactivate receptors until new ones are synthesized, lowering Emax (S22, S22 notes).
- **E.** A partial agonist itself gives submaximal activation (S22); it is not described as a blocker overcome by more agonist.

Q8 · Clinical vignette · Pharmacodynamics: Targets & Receptors

A 54-year-old is given phenoxybenzamine before surgery for pheochromocytoma. Blood pressure stays controlled even when the tumor releases large catecholamine surges. What best explains why the blockade persists?

A It activates opposing receptors, producing a physiologic counter-effect instead

B It binds receptors irreversibly; effect lasts until new receptors are made

C It is a partial agonist that caps the maximal catecholamine response

D It neutralizes catecholamines chemically in plasma before they reach receptors

E It competes reversibly, so high catecholamine levels easily displace it

ANSWER: B

An irreversible antagonist's duration depends on receptor turnover, and high agonist levels cannot overcome it (p. 25). p. 25-27

- **A.** That is physiologic antagonism, like insulin vs glucocorticoids (p. 27).
- **C.** Partial agonists produce a submaximal response; phenoxybenzamine is an irreversible antagonist (p. 25; p. 26).
- **D.** That is chemical antagonism, like protamine–heparin (p. 26).
- **E.** That describes competitive antagonism, which surges CAN overcome (p. 25).

Q9 · Mechanism · Pharmacodynamics: Targets & Receptors

A patient asks why an inhaled bronchodilator works in minutes but an oral corticosteroid takes hours. What best explains the corticosteroid's delay?

- **A** It acts on an intracellular receptor that changes gene transcription
- **B** It is a partial agonist with low efficacy
- **C** It acts on a ligand-gated ion channel
- **D** It must first be activated by CYP2D6
- **E** It binds a G protein–coupled receptor that raises cAMP

ANSWER: A

Gene-acting drugs need time to make new proteins, so effects lag 30 minutes to several hours (p. 28). p. 28

- **B.** Efficacy does not explain timing (p. 28).
- **C.** Ion channels respond within milliseconds (p. 31).
- **D.** Prodrug activation is not the reason for this lag (p. 28).
- **E.** That is how a β_2 agonist bronchodilator works, quickly (p. 31; p. 32).

Q10 · Mechanism · Pharmacodynamics: Targets & Receptors

A drug activates a receptor coupled to Gq. Which change follows inside the cell?

- **A** Adenylyl cyclase is stimulated, so cAMP rises
- **B** Phospholipase C makes IP_3 and DAG; calcium rises
- **C** Phosphodiesterase breaks down cGMP in the cell
- **D** A JAK kinase activates STAT transcription proteins
- **E** Adenylyl cyclase is inhibited, so cAMP falls

ANSWER: B

Gq stimulates phospholipase C, producing IP_3 (calcium release) and DAG (protein kinase C) (p. 32; p. 35). p. 32; p. 35

- **A.** That is Gs (p. 32).
- **C.** That is Gt in the retina (p. 32).
- **D.** That is a cytokine receptor, not a G protein (p. 29).
- **E.** That is Gi (p. 32).

Q11 · Clinical vignette · Dose-Response Relationships

A PA is starting a patient on lithium. Compared with a drug that has a wide therapeutic index, what does lithium's narrow therapeutic index mean for this patient's care?

- **A** Lithium is more potent, so it can be dosed without monitoring levels
- **B** Lithium has a higher Emax, so lower doses than usual are always enough
- **C** Lithium is 100% bioavailable, so blood levels never need to be checked
- **D** Effective and toxic doses are close, so dose carefully and monitor blood levels
- **E** Lithium has a large safety margin, so dosing errors are well tolerated

ANSWER: D

A narrow TI means effective and toxic doses are close together — requiring careful dosing and often blood-level monitoring; lithium is a classic narrow-TI drug (S38). S38; S38 (notes)

- **A.** Potency matters for dosing amount, not safety margin; narrow-TI drugs need monitoring (S34, S38).
- **B.** TI is about toxic vs. effective doses, not Emax (S30, S38).
- **C.** Bioavailability is a different concept (S50); narrow-TI drugs need monitoring (S38).
- **E.** That describes a WIDE TI (S38).

Q12 · Clinical vignette · Dose-Response Relationships

Two drugs relieve the same symptom. Drug A works at 5 mg and Drug B at 50 mg, but at maximal doses Drug B produces a much greater maximal effect. Which statement best guides drug selection?

A Drug B has greater efficacy, which matters more clinically than Drug A's potency

B Drug A should be chosen because greater potency means it works better

C Both drugs have equal efficacy because both relieve the same symptom

D Drug A has greater efficacy because it needs a much smaller dose

E Drug B has greater potency because its maximal effect is much higher

ANSWER: A

Efficacy (Emax) clinically matters much more than potency — a highly potent drug with low efficacy may be clinically useless (S34). S30; S34; S34 (notes)

- **B.** Potency alone never tells you whether a drug works better (S34 notes).
- **C.** Drug B reaches a higher maximal effect, so its efficacy is greater (S30).
- **D.** A smaller dose means greater POTENCY; efficacy is the maximum effect (S30).
- **E.** Higher Emax = more efficacious, not more potent (S34).

Q13 · Clinical vignette · Pharmacokinetics: ADME — Absorption & Distribution

A patient stable on IV propranolol is switched to the oral form and needs a much higher oral dose to reach the same systemic exposure. Which best explains this?

- **A** Propranolol has a narrow therapeutic index, so oral doses must be higher
- **B** Extensive first-pass hepatic metabolism lowers the drug's oral bioavailability
- **C** Propranolol is highly protein-bound, so the oral form is inactive in plasma
- **D** The IV form was less than 100% bioavailable, so less drug reached the blood
- **E** Oral propranolol cannot cross the blood-brain barrier, so more must be given

ANSWER: B

Propranolol has a large first-pass effect; oral doses pass GI → portal → liver before systemic circulation, so higher oral doses are needed to match IV exposure (S51, S50). S50; S51

- **A.** Narrow TI calls for careful dosing and monitoring (S38); propranolol is named for first-pass, not narrow TI (S51).
- **C.** Protein binding affects free drug in plasma regardless of route (S52); the IV-to-oral gap is the first-pass effect (S51).
- **D.** IV has 100% bioavailability by definition (S50).
- **E.** BBB crossing relates to CNS entry (S53), not systemic exposure after an oral dose.

Q14 · Clinical vignette · Pharmacokinetics: ADME — Absorption & Distribution

A patient with meningitis receives a highly polar antibiotic. Which statement best describes this drug's ability to reach the CNS?

- **A** Polar drugs never reach the CNS, even when the meninges are inflamed
- **B** Polar drugs cross easily because P-glycoprotein pumps them into the CNS
- **C** Polar drugs usually cross poorly, but inflamed meninges allow better entry
- **D** Polar drugs cross easily because tight junctions favor polar molecules
- **E** Polar drugs cross well because the brain is a poorly perfused tissue

ANSWER: C

A highly polar antibiotic generally would not cross the BBB well — unless the meninges are inflamed, as in meningitis (S53 notes). S53; S53 (notes)

- **A.** Inflamed meninges (meningitis) are the stated exception (S53 notes).
- **B.** Efflux transporters like P-glycoprotein pump some drugs back OUT of the CNS (S53).
- **D.** Tight junctions restrict paracellular movement; small lipophilic non-ionized drugs cross more easily (S53).
- **E.** The brain is highly perfused (S52), and perfusion does not overcome the BBB for polar drugs (S53).

Q15 · Clinical vignette · Pharmacokinetics: ADME — Absorption & Distribution

A patient overdoses on a weak-acid drug. To speed its renal excretion, what should be done to the urine?

- **A** Make the urine more acidic
- **B** Slow glomerular filtration
- **C** Keep urine pH unchanged and give more fluids only
- **D** Raise plasma protein binding
- **E** Make the urine more alkaline

ANSWER: E

In alkaline urine a weak acid is mostly ionized, so it cannot diffuse back across the tubule and is trapped in the urine (p. 10). p. 10

- **A.** Acidic urine keeps a weak acid nonionized and lipid-soluble, so more is reabsorbed (p. 10).
- **B.** Less filtration would slow excretion, not speed it (p. 10).
- **C.** Fluids alone do not trap the drug; ionization is the lever (p. 10).
- **D.** Protein binding does not trap drug in the urine (p. 8; p. 10).

Objectives checklist

Tick only what you could teach out loud. Saved on this device.

- [ ] **IO1.1** Define and describe fundamental pharmacologic concepts, including pharmacokinetics, pharmacodynamics, drug-receptor interactions, dose-response relationships, therapeutic index, half-life, and steady state.
- [ ] **IO1.2** Describe the processes of drug absorption, distribution, metabolism, and excretion and factors that influence these processes.
- [ ] **IO1.3** Explain how patient-specific factors, including age, pregnancy, renal and hepatic function, and pharmacogenetic variation, may affect medication response.
- [ ] **IO1.4** Identify common types and mechanisms of adverse drug reactions, drug-drug interactions, and drug-disease interactions.
- [ ] **IO1.5** Interpret common components of medication orders and prescriptions, including drug name, dose, route, frequency, and duration.

- [ ] **SO1** Define pharmacology, pharmacodynamics, and pharmacokinetics.
- [ ] **SO2** Describe the major stages of drug development and FDA approval.
- [ ] **SO3** Explain how drugs interact with receptors to produce an effect.
- [ ] **SO4** Interpret a dose-response curve, including potency and efficacy.
- [ ] **SO5** Explain absorption, distribution, metabolism, and excretion (ADME), including bioavailability and the first-pass effect.